Hybrid Weather Forecasting: A Multi-Source Data Fusion Approach Using Multiple APIs and Machine Learning

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***Abstract*—This project aims to improve short-term weather forecasting, with the primary objective of combining multiple sources of data from the OpenWeather API (ground weather observations) and the NASA POWER API (satellite-based measurements). We also aligned and cleaned the hourly weather data, after which we performed exploratory data analysis (EDA) to look at variable distributions, correlations, and importance of features. The combination of data sets expanded the feature space and helped diminish both error and variability in the predictions. Lag features were engineered before developing two long-short-term memory (LSTM) models to predict temperature and humidity by taking in sequences of 24 hours of sequence of weather data and predicting the weather for the next hour. The experimental results indicate that there is more value in using both satellite and ground data for a more holistic perspective and presenting more reliable weather forecasts.**

***Index Terms*—Weather forecasting, data fusion, LSTM, multisource APIs, machine learning**

I. Introduction

Weather plays a critical role in several fields. These fields include agriculture, disaster, transport, and energy distribution. These forecasting models mostly use data from one source and their accuracy is compromised when data sharing is limited or uneven across an area. In general, the main aim of this research is to design a hybrid weather forecast framework using weather data from multiple sources. The hybrid system enriches the input of the value in spatial and temporal dimensions by combining the past records of public APIs with the measurements of satellite, ground-based, air-borne and space-borne sensors[1]. Since meteorological variables are sequential in nature, long-short-term memory (LSTM) neural networks are used to predict major climate parameters such as climate temperature, humidity, and wind speed with relatively high accuracy. This multisource machine learning-based forecasting approach aims to magnify forecast accuracy and thus enable the facilitation of decision-making from the object-reality basis[2].

II. Problem Statement

Usually, traditional forecast systems stand with limited accuracy by being generally based on one data source, not mapping well with spatial or temporal variability in atmosphere. These constraints become more poignant in areas that have very few sensors or in areas where records are not maintained with unanimity historically. As the other variables grow complex along with the growing data availability on a multitude of platforms, it becomes ever more pressing to turn to forecasting models capable of fusing heterogeneous data sources. The research tries to address the problem of improving weather prediction accuracies by proposing a hybrid forecasting framework that merges data from various sources such as public weather APIs, satellite observations, or remote sensing datasets and uses the Long Short-Term Memory (LSTM) neural networks to model the temporal dependence[3]. This research faces challenges in adapting data formats for various sources resolving inconsistencies and training an efficient sequence model able to provide reliable and accurate forecasts for some primary meteorological parameters.

III. Review of Literature

A. Suleman & Shridevi (2022)[2]

In an attempt to advance short-term air temperature forecasting, the authors presented an improved LSTM model augmented by spatial attention mechanisms. This allowed the model to be aware of spatial variability and locally influenced weather phenomena, leading to greater prediction accuracy.

B. Musah et al. (2022)[3]

In this study they mentioned, it was assessed whether the data from the OpenWeatherMap API were as reliable as those from INMET, the national weather service of Brazil. The results revealed that the data from the API were sometimes inconsistent and must be validated independently before integration into a larger forecasting system.

C. Mohammed et al. (2024)[4]

This work features enhancements in weather forecasting that benefit from data fusion by putting together different machine learning models with weather and AQI data. The results further confirmed that the greater the sources of data one can combine, the better the forecast.

D. Bandara et al. (2020)[5]

This paper proposes LSTM-MSNet, an LSTM-based network for multiple seasonal pattern modeling for time-series forecasting. It has served multiple applications for meteorological datasets with seasonality.

E. Liu et al. (2024)[6]

This research put forth STFM, a spatiotemporal forecasting model to overcome the geostatistical accuracy problems in weather prediction. The approach on the spatial and temporal sides can outdoors support motivation of our hybrid approach.

F. Zhang et al. (2021)[7]

MFSP-Net, a recurrent neural network for short-term precipitation nowcasting, was introduced in this study. Radar, reanalysis, grid datasets, with emphasis on multimodal data fusion, are the very essence of the approach pursued by this project.

G. Wang et al. (2023)[8]

TAFFNet adaptively fuses radar data with numerical weather prediction outputs. This model represents an intent to integrate various weather datasets into a single deep learning framework.

H. Karevan & Suykens (2018)[9]

The two-layered stacked LSTM introduced in the study captures spatial-temporal dependencies and performs temperature prediction across different geographical locations.

I. Kim et al. (2017)[10]

DeepRain, as introduced, is a ConvLSTM-based model that predicts precipitation by way of multichannel radar data. The architecture of DeepRain showed enhanced accuracy compared to conventional models.

J. Al-Hajj et al. (2021)[11]

This research combined LSTMs and genetic programming to work on solar radiation prediction. The hybrid model enabled accurate results and emphasized the advantages of integrating approaches to learning.

K. Abdellaoui & Mehrkanoon (2020)[12]

The authors proposed an ensemble explainable deep learning model that enhance forecasts with data collected from multiple weather stations and made the predictions of the model interpretable.

L. Patel et al. (2018)[13]

The short-range precipitation forecasting problem was solved here with a hybrid LSTM and CNN model. It captured temporal features and long-term dependencies well.

M. Tekin et al. (2024)[14]

The authors extended ConvLSTM by introducing attention and context matcher mechanisms. Their architecture achieved enhancement in accuracy for numerical weather prediction tasks.

IV. Objectives

A. Collecting and combining data from multiple weather data sources

Collect and combine both current and past weather data from a number of APIs and data sets (for example, DFW Weather, OpenWeatherMap) in order to provide weather models with a range of input data and sources.

B. Performing historical weather trend analysis

Detect historical weather trends by evaluating the level of historic weather patterns, seasonal or otherwise, as well as anomalies or other factors that may help build validation cases that create models able to learn from a greater variety of potential patterns of highly reliable forecasts.

C. Implementing data embedding and fusion techniques

Explore the application of data embeddings[4], and fuse feature sets from many data sources in order to record and normalise both temporal and spatial associations you find between different weather datasets.

D. Developing hybrid weather prediction models

Build a better hybrid weather forecasting architecture that takes LSTM and combines this with some other predictive engine like spatial attention to enhance the models prediction accuracy when weather conditions are complex with many dependencies upon themselves.

E. Evaluating model performances

Use evaluation metrics including RMSE, MAE, and R to evaluate the performance of your models, comparing model outputs with actual observed weather conditions.

V. Data Collection and Preprocessing

A. OpenWeather API

Initially, the primary research dataset was obtained from OpenWeather, being downloaded from their official bulk historical weather data service. Subscribing to the plan called History Bulk, designed for long-range historical weather data, the researchers got hourly weather records for over 46 years, ranging from 1st January 1979 until now. As a benefit of the subscription, the researcher is entitled to retrieve 15 crucial weather parameters such as temperature, relative humidity, wind speeds, atmospheric pressure, cloud cover, and so forth.

A single geographic location was chosen in this study, being the Dallas-Fort Worth International Airport, on account of ensuring consistency of data quality and resolution. The OpenWeather platform enables a user to fix a location by name or deflect coordinates themselves around an interactive map; the required date range can also be fixed and and choose the output file type [json, csv] at the time of placing the request.

B. NASA POWER API

After procuring the first dataset from OpenWeather API, other meteorological parameters were to be obtained from the NASA POWER API. The approach was to create a parameterized URL and test it with Postman. The URL format shared below is the endpoint used generally: ”https://power.larc.nasa.gov/api/temporal/hourly/point? parameters=WS2M,ALLSKY_SFC_LW_DWN,RH2M,PS, ALLSKY_SFC_SW_DWN,T2M,PRECTOTCORR &community=RE&longitude=xxxx&latitude=xxxxx&start=xxxxxx &end=xxxxxx&format=CSV&header=true&time-standard=UTC”.

This URL can be customized by giving different longitude, latitude, start date, and end date as parameters. Parameters extracted included 2-meter wind speed (WS2M), downward longwave radiation (ALLSKY-SFC-LW-DWN), surface pressure (PS), relative humidity (RH2M), surface shortwave radiation (ALLSKY-SFC-SW-DWN), temperature (T2M), and precipitation (PRECTOTCORR) for a given place.

The API caters to output either in CSV file format or JSON file format, according to the needs of the user. For further ease in handling and later integration with other datasets, CSV format was chosen. The resulting files were downloaded and made ready to be metamorphosed and merged with the OpenWeather data.

As mentioned in the official documentation downloadable at the NASA POWER API website[1], this parameterized URL can be constructed programmatically or interactively. The NASA Data Access Viewer (DAV) tool[2] was also used to simplify the generation of valid URLs by selecting parameters, coordinates, and time ranges through a graphical interface.

[1]https://power.larc.nasa.gov/docs/services/api/

[2]https://power.larc.nasa.gov/data-access-viewer/

VI. Exploratory data analysis

Exploratory data analysis is generally carried out to study the structure, distribution, and relationships that exist between different weather variables prior to model building. This analysis is done on two major datasets: the OpenWeather API dataset, which provides ground readings of weather parameters, and the NASA POWER API dataset, which provides atmospheric parameters as derived by satellites. The two datasets were collected at an hourly frequency and covered concurrent time periods and were thus compatible for integration and comparison. We began with cleaning and prepping data. The datetime field of the two data sets was parsed and normalized such that their formats are alike, allowing appropriate temporal alignment on merge. In the two datasets also, units were normalized-for example, temperature units were converted to Celsius, wind speed was measured in meters per second-to account for possible inconsistencies because those could pose serious problems for feature scaling and model performance. Descriptive statistical analysis was also employed to detect variation and distribution of major weather variables, such as temperature, relative humidity, dew point, pressure, and wind speed. Summary statistics provided information on central tendencies and dispersion of all the attributes. Correlation analysis indicated strong positive correlations of temperature and dew point and a moderate negative correlation of temperature and humidity, according to established meteorological principles. [1] Notably, the NASA POWER dataset exhibited less variability than the OpenWeather data, as would be expected by virtue of satellite-based averaging smoothing out. To guarantee the effectiveness of data fusion, we investigated the feature space of the fused dataset. The fused dataset presented a more detailed representation of weather conditions by incorporating complementary features from both datasets. We also conducted a feature importance analysis using mutual information scores, which confirmed that incorporating NASA POWER’s variables improved the predictive capability of the fused dataset. This further supported our hypothesis that fusion of multi-source data had the potential to produce more stable and reliable weather forecasting models.

VII. Hypotheses for the Study

From our preliminary background research and exploratory analysis, we expect integration of multiple weather data sources to enhance forecast performance. In particular, we speculate that data fusion combining ground-based measurements in the OpenWeather API with satellite measurements in the NASA POWER API will enhance forecasting precision more than individual use of each dataset. The rationale behind this hypothesis is that multisource data provides a richer and higher-fidelity view of weather, providing the LSTM model with richer signals on which to learn. Merging the strengths of ground and satellite observations, we hope that the model will better capture temporal patterns and dependencies and eventually result in more accurate short-term weather forecasting[1][3].

VIII. Data Analytics

Our analytics data pipeline was created to transform raw multi-source weather data into actionable intelligence and accurate short-term predictions using deep learning models. The process involved multiple steps, from data preprocessing and data integration to feature engineering, model creation, and model evaluation. We began with data preprocessing and fusion by collecting hourly weather data from two sources OpenWeather, which logs ground-level measurements, and NASA POWER, which provides satellite-derived atmospheric measurements. The two datasets were merged on a common datetime index for time alignment. Standardization was performed to unify the units of measurement -for instance, recording temperature readings in Celsius and performing statical method for merging the data [4]. The two datasets were merged on the timestamp through a left join into a singular, enriched dataset containing complementary weather attributes. Important weather parameters such as temperature, humidity, dew point, wind speed, and atmospheric pressure were selected for feature engineering toward time series modeling. Lag features were created to represent 24 hours of previous weather data, to allow the model to learn time dependencies and trends that would help with common forecasting tasks. For modeling, two separate LSTM networks were trained one to predict temperature and the other one for humidity; both types of models accept 24 sequences of hourly readings as input to produce the predicted value for the next hour. We used the sliding window approach for producing such composed input sequences across the whole dataset. Both models were trained using the Adam optimizer with a learning rate of 0.0001, and were optimized using MSE (Mean Squared Error) loss-function working with its prime objective, to minimize prediction error and maximize the accuracy of the results.

IX. Data Visualization and Results Report

The below are the results which we got for Temperature and humidity with respect to LSTM with OpenWeather and LSTM with OpenWeather and NASAPOWER.We significantly increased prediction accuracy through merging datasets from multiple sources to produce a more accurate and exact visual representation of weather conditions.LSTM gave better predictions on the fused dataset, reducing errors and improving R scores regarding the prediction of humidity and temperature, thus emphasizing how well data fusion works for accuracy in forecasting and other considerations of importance in integrating multi-information sources for an accurate and efficient environmental treatment technique[2].

A. Temperature

We have calculated MSE, MAE, and R-Squared, values that indicate how accurate forecasted weather can be when only OpenWeather data is used or when Open- Weather and NASA POWER data are combined. With a higher coefficient of correlation and reduced errors of MSE and MAE, the whole information set performs much better in a way. This means that predictions based on both data sources vastly outstrip accuracy, with the closest to actual weather conditions. Using both NASA POWER and OpenWeather, therefore, massively increases the overall reliability of the simulator.

LSTM with OpenWeather: This graph explains the expected and actual outcomes of a dataset, most likely one that is associated with a particular time sequence. The line in orange displays the actual values, while the blue line depicts the model's test predictions. The fact that the two lines are very similar to one another suggests that the model has performed well in forecasting the values. The predictions system's highly precise and reliability are indicated by the small space along the diagonal lines.

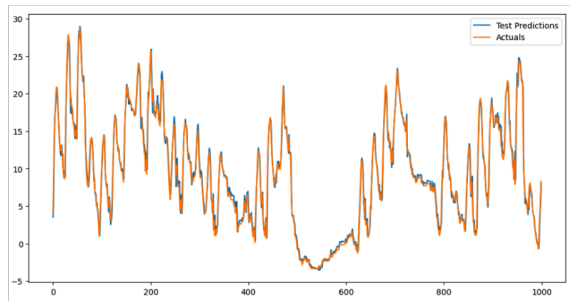

Fig. 1. Temperature Curve with OpenWeather

LSTM with OpenWeather and NASAPOWER: This graph is a comparison of anticipated values in blue and actual values in orange with respect to a data set or its series or period. The two lines track one another very close to one another. Thus, it shows how well the forecasting model performs. Since the distinguished separation became a little more, the algorithm probably already had a good understanding of the data pattern. More or less clutch to this is the mark of higher R2 scores and smaller error values, thereby measuring a high accuracy of the predictions.

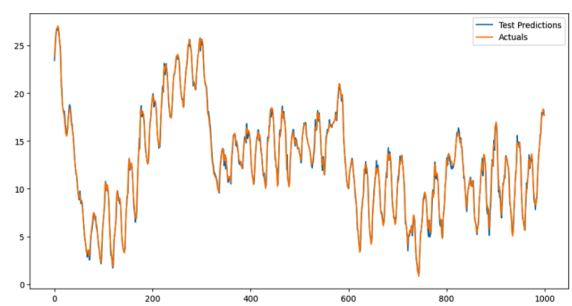

Fig. 2. Temperature Curve with OpenWeather and NASAPower

Metric	OpenWeather	OpenWeather + NASA POWER
MSE	0.9059	0.2441
MAE	0.6863	0.3728
R^2	0.9911	0.9971

TABLE I
Comparison of forecasting metrics for OpenWeather vs. OpenWeather + NASA POWER

B. Humidity

The forecasts had turned much better with information from the two sources combined, namely, OpenWeather and NASA POWER: the Mean Squared Error (MSE) registered a 92 percent fall, from 111.37 to 8.94, while the Mean Absolute Error (MAE) experienced a 76 percent reduction, from 8.96 to 2.11, ensuring that prediction performance was greater. The R value upwardly shifted from 0.6018 to 0.9564, further supporting the fact that combining the datasets enabled the models to better explain weather data's variance.

Metric	OpenWeather	OpenWeather + NASA POWER
MSE	111.3719	8.9413
MAE	8.9569	2.1138
R^2	0.6018	0.9564

TABLE II
COMPARISON OF FORECASTING METRICS FOR OPENWEATHER VS. OPENWEATHER + NASA POWER

LSTM with OpenWeather: This suggests that LSTM model trained by the data from OpenWeather like also included weather-related features.The test line was the blue colored line which represents values which are being predicted by the LSTM model on a set of unseen data which is known as the test set.Also,the actuals were in Orange line which represents the true observed values for the same time period as the test predictions. Comparing the blue and orange lines, we could how well the LSTM model performed. In this graph, we could see that the model captures the overall trends and cyclical patterns of the actual data. There are periods where the predictions deviate from the actual values particularly around the middle section of the graph where the blue line seems to flatten out while the orange line fluctuates more.

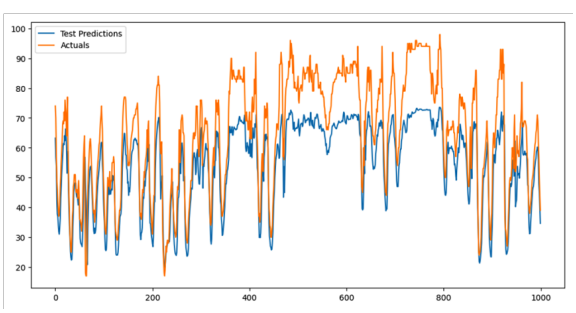

Fig. 3. Humidity Curve with OpenWeather

LSTM with OpenWeather and NASAPOWER: This indicates that this LSTM model was trained using data from both OpenWeather and NASAPOWER. NASAPOWER provides access to NASA's climate and weather datasets this model is incorporated broader range of variables. Also, Test Predictions in blue line show the model's predictions on the test set. The orange-colored line denotes the actuals, which means these values are still ideas to be observed in real life for the same period. Deviations of any sort from the predicted values are smaller and fewer. This gives us conclusively good proof that adding data from NASAPOWER along with OpenWeather data improved the LSTM model a lot in forecasting the target variable. The model could then describe the interactions and fluctuations in the actual data.

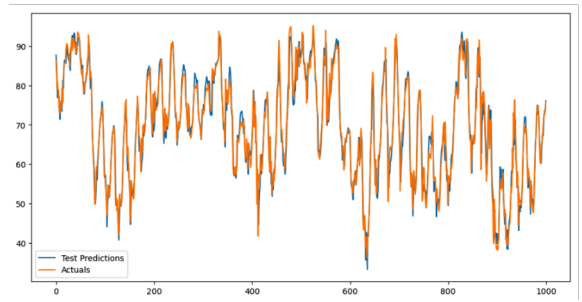

Fig. 4. Humidity Curve with OpenWeather and NASAPower

X. CONCLUSION

We found that the inclusion of ground-based OpenWeather data, along with satellite-derived NASA POWER data, greatly improved the predictive accuracy of our weather forecast model. After careful data pretreatment, we created a comprehensive data set that included a range of atmospheric characteristics (including temperature, humidity, etc.) through temporal alignment, unit conversion. During the exploratory analysis, our insights included recommendations on the variety of available atmospheric characteristics and how strongly temperature was correlated with the dew point. We also identified the variability within each source of data. Key features were engineered within the dataset, with a distinct emphasis on lag variables, to provide our LSTM models the opportunity to learn exactly how accurately capturing chemical interactions can lead to more stable temporal patterns of the connected atmosphere. Training two different models, one for temperature prediction and one for humidity predictions, we demonstrated that a single relational fused data set could provide not only better temperature and humidity prediction, but more consistent model stability. Improved model performance also affirms our hypothesis that data fusion creates better spatial predictive accuracies by being able to capture more complement characteristics of weather systems. Overall, this study provides support for the use of multisource data sources in environmental modeling and provides a recommendation for an easily-utilized scalable pipeline to engage and/or accelerate such data metrics and other types of forecasting in deep learning settings. The resulting images from our recommendations are transferable and extend beyond this audience, meaning that this area of research could be looked at in the more general field of climate analytics.

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